

Polynomial-time decision of local-unitary equivalence for graph states

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Abstract

We give a deterministic polynomial-time algorithm for deciding local-unitary equivalence of labelled graph states. The starting point is the quasi-polynomial algorithm of Claudet and Perdrix, whose expensive step lists high-order incidence constraints for generalized local complementations. We show that the same row module over $\mathbb{Z}/2^r\mathbb{Z}$ can be generated from pair and triple rows by a polynomial number of diagonal-map and Hermite-normal-form operations. The closure takes at most $\max\{0, r - 2\}$ enlargement steps. An integer dual lattice then gives the complete modular kernel and its binary space of edge toggles, together with witnessing multiplicities. Substituting this computation into the published graph reduction gives polynomial bit complexity. We prove that the auxiliary graphs satisfy the constrained local-Clifford algorithm’s hypotheses and handle disconnected inputs and zero toggle spaces separately.

1 Introduction

A graph state associates an n -qubit pure state with a finite simple graph. Its graph representation makes local-Clifford transformations amenable to discrete methods, and local-Clifford equivalence can be decided efficiently [3]. Local-unitary equivalence allows arbitrary single-qubit unitaries and is a different decision problem. Throughout this paper the vertices are labelled: the algorithm does not optimize over qubit permutations.

Claudet and Perdrix [1] reduced local-unitary equivalence to a sequence of graph and linear-algebra computations and obtained the bound $n^{\log_2 n + O(1)}$. Their standard-form reduction leaves one generalized local complementation, together with ordinary local complementations on a specified set of vertices. The generalized operation is described by linear congruences, but the direct description has a row for every relevant high-order subset. At level r , this creates $O(n^{r+1})$ rows. The level required for the general local-unitary question is logarithmic in n . Whether the general decision problem admits a polynomial-time algorithm is also explicitly posed in Claudet’s thesis [2, Chapter 8].

The subset enumeration can be avoided. Except at the pair-to-triple transition, adjoining a vertex to a subset acts on its weighted incidence row by a fixed diagonal linear map. Starting with both pair and triple rows therefore suffices. Repeatedly applying these maps, and reducing the row lattice to Hermite form, computes all constraints without retaining the subsets that index them. Only $r - 2$ enlargement steps can be needed when $r \geq 2$.

Theorem 1.1 (Main result). *There is a deterministic classical algorithm that, given two finite simple graphs G and H on the same labelled vertex set of size $n \geq 1$, decides in polynomial bit complexity whether their graph states are local-unitary equivalent, up to global phase. No connectivity, degree, or graph-class promise is required.*

The graph-theoretic reductions are due to Claudet and Perdrix. The new ingredient is the polynomial computation of the incidence module; the complete toggle space then follows by modular linear algebra. Sections 3 and 4 give these constructions. Sections 5 and 6 explain their use in the published reduction, including the hypotheses needed for its constrained local-Clifford step. Section 7 reports exact diagnostic checks and their limits.

2 Problem and notation

For a finite simple graph $G = (V, E)$ with $V = [n]$, define

$$|G\rangle = \prod_{\{u,v\} \in E} \text{CZ}_{uv} |+\rangle^{\otimes n}, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (1)$$

Local-unitary equivalence means that

$$|H\rangle = e^{i\phi} \left(\bigotimes_{v=1}^n U_v \right) |G\rangle \quad \text{for some } U_v \in U(2), \phi \in \mathbb{R}. \quad (2)$$

The labels in (2) are fixed. A local complementation at v toggles every edge among distinct neighbors of v . We write $G \star v$ for the resulting graph and use LC for equivalence under sequences of these operations.

For a multiset S of vertices, $s(x)$ denotes its nonnegative integer multiplicity and $\text{supp}(S) = \{x : s(x) > 0\}$. For $K \subseteq V$, let

$$c_S(K) = \sum_{x: K \subseteq N_G(x)} s(x) \quad (3)$$

be its common-neighbor multiplicity. The following conventions are those of [1, Definitions 2.1–2.2].

Definition 2.1. Let $r \geq 1$. An independent multiset S is r -incident if, for every $0 \leq j < r$ and every $K \subseteq V \setminus \text{supp}(S)$ with $|K| = j + 2$,

$$c_S(K) \equiv 0 \pmod{2^{r-j-\delta_{j0}}}. \quad (4)$$

Its r -local complementation toggles a pair outside $\text{supp}(S)$ precisely when its common-neighbor multiplicity is 2^{r-1} modulo 2^r . Pairs meeting the support are unchanged.

Let LC_r denote the local-equivalence relation generated by single-qubit Hadamards and $Z(\pi/2^r)$ rotations, in the convention of [1]. Thus $\text{LC}_1 = \text{LC}$, and every r -local complementation is a local-unitary transformation. The equivalence between the graph operations and these unitary relations is established in [1, Section 2.3].

Our input graphs have adjacency-matrix encodings. Arithmetic complexity means bit complexity. The auxiliary parameter r is charged according to its value, not merely its binary encoding length. This is sufficient because the final algorithm chooses $r = O(\log n)$. The ring $R = \mathbb{Z}/q\mathbb{Z}$, where $q = 2^r$, is not a field when $r > 1$; its kernels need not be free modules.

3 Compression of the incidence module

We first state the algebraic result independently of graph standard forms. Let $a_1, \dots, a_t \in \{0, 1\}^m$. For $K \subseteq [t]$, define the coordinatewise product

$$a_K = \bigodot_{v \in K} a_v, \quad w(k) = \begin{cases} 1, & k = 2, \\ k - 2, & k \geq 3. \end{cases} \quad (5)$$

The weighted incidence module is

$$M_r = \text{span}_R \{2^{w(|K|)} a_K : 2 \leq |K| \leq r + 1\} \subseteq R^m. \quad (6)$$

Subsets larger than t contribute no rows.

Theorem 3.1 (Polynomial incidence compression). *A generating matrix for M_r with at most m rows can be constructed deterministically in time polynomial in m, t, r . The construction uses at most*

$$B(r, t) = \max\{0, \min\{r - 2, t - 3\}\} \quad (7)$$

strict lattice enlargements, and its retained integer basis entries have $O(r + 1)$ bits.

3.1 The closure identity

For each $v \in [t]$, define the R -linear map

$$T_v(z) = 2a_v \odot z. \quad (8)$$

Let N_r be the smallest submodule containing $2a_K$ for $|K| = 2$ or 3 and invariant under all the maps T_v .

Lemma 3.2. $M_r = N_r$.

Proof. The pair seeds occur in (6). The triple seeds occur when $r \geq 2$; when $r = 1$, all seeds are zero modulo two. Thus every seed belongs to M_r .

We check invariance on a generating row $2^{w(k)} a_K$, where $k = |K|$. If $v \in K$, the binary coordinates satisfy $a_v \odot a_K = a_K$, so T_v gives twice that row. If $v \notin K$ and $k = 2$, the result is $4a_{K \cup \{v\}}$, twice the triple seed. For $r = 1$ this result is zero. If $v \notin K$ and $k \geq 3$, then

$$T_v(2^{k-2} a_K) = 2^{k-1} a_{K \cup \{v\}}, \quad (9)$$

which is the next prescribed row, unless $k = r + 1$, in which case it vanishes modulo 2^r . Hence M_r is invariant and $N_r \subseteq M_r$.

Conversely, if $|K| = k \geq 4$, choose three vertices of K and start with their triple seed. Applying T_v for the remaining $k - 3$ distinct vertices gives coefficient $2 \cdot 2^{k-3} = 2^{k-2}$. Every generator in (6) is therefore in N_r . Pair and triple generators were already seeds. This proves equality; in particular, repeated application of a vertex introduces no extraneous constraints. $\square$

3.2 A bounded lattice computation

Proof of Theorem 3.1. The case $m = 0$ is immediate, so assume $m > 0$. Lift the seed rows to integers and let $L_0 \subseteq \mathbb{Z}^m$ be the lattice generated by those rows and $q\mathbb{Z}^m$. Extend (8) to the same integer diagonal map and iterate

$$L_{i+1} = L_i + \sum_{v=1}^t T_v(L_i). \quad (10)$$

After each step, retain a full-rank row Hermite basis. Applying T_v to a basis suffices to generate $T_v(L_i)$.

The image of every L_i modulo q is contained in M_r , by the invariance proved in Lemma 3.2. Conversely, a prescribed row indexed by a set of size $k \geq 4$ is obtained from a triple seed after $k - 3$ applications of the maps. The largest relevant size is $\min\{t, r + 1\}$. Thus every prescribed row is present by step $B(r, t)$; when $r \leq 2$ or $t \leq 3$, the seeds already suffice. Since each lattice contains $q\mathbb{Z}^m$, equality of their images modulo q is equality of the lattices themselves. Hence $L_{B(r,t)}$ is stable. If equality occurs earlier, invariance prevents any later change. A further Hermite computation detects stability.

The initial matrix has at most $m + \binom{t}{2} + \binom{t}{3}$ rows. Every subsequent matrix has at most $m(1 + t)$ rows. A reduced triangular Hermite basis of a lattice containing $q\mathbb{Z}^m$ has each diagonal entry dividing q : reduce the vectors qe_j against the triangular basis successively, using the orientation corresponding to that basis. Off-diagonal entries are reduced below the corresponding positive pivots. Consequently every retained entry has magnitude at most q and has $O(r + 1)$ bits. Applying T_v multiplies selected coordinates by two, preserving this bit bound up to one bit.

Deterministic Hermite-normal-form algorithms have polynomial bit complexity in their matrix dimensions and input bit lengths [4]. Polynomially many calls on the bounded matrices above establish the asserted bit complexity, including intermediate arithmetic inside each call. Reducing the final m basis rows modulo q gives a generating matrix B for M_r . Zero rows may be omitted. $\square$

Algorithm 1: weighted incidence compression

1. Form the pair and triple rows $2a_K$ and the rows of qI_m .
2. Replace their integer row span by a canonical Hermite basis H .
3. Compute a Hermite basis H' of the rows of H and all $T_v(H)$.
4. If $H' = H$, return $H \bmod q$; otherwise set $H \leftarrow H'$ and repeat step 3.

For $m = 0$, return the empty matrix immediately.

Corollary 3.3. *For $s \in R^m$, the exponentially indexed conditions*

$$2^{w(|K|)} a_K \cdot s = 0 \pmod{q} \quad (2 \leq |K| \leq r + 1) \quad (11)$$

hold if and only if $Bs = 0$ in R .

Proof. Annihilating generators of a row module is equivalent to annihilating the entire module. Apply Theorem 3.1. $\square$

4 The complete toggle space and its witnesses

Fix a graph G , an independent set $X \subseteq V$, and $Y = V \setminus X$. Put $m = |X|$, $q = 2^r$, and $h = 2^{r-1}$. For each $v \in Y$, let $a_v \in \{0, 1\}^X$ be its adjacency vector into X . Let B be the compressed matrix and

$$\Sigma = \ker_R B. \quad (12)$$

4.1 Matching the precise support convention

Lemma 4.1. *The vectors in Σ , represented by multiplicities in $\{0, \dots, q - 1\}$ on X , are exactly the r -incident multisets supported in X .*

Proof. For $K \subseteq Y$, the common-neighbor multiplicity is $a_K \cdot s$. When $k = |K|$, condition (4) requires divisibility by $2^{r-w(k)}$, exactly equivalent to (11). This equivalence also holds for $r = 1$, when the pair condition is vacuous.

The actual support of a multiplicity vector may be smaller than X . If a set $K \subseteq V \setminus \text{supp}(S)$ contains some $x \in X$, no vertex of X is adjacent to every vertex of K : independence of X forbids adjacency to x . Its common-neighbor multiplicity in S is therefore zero. All such extra conditions in Definition 2.1 hold automatically. These observations establish both directions. Reducing multiplicities modulo q preserves all relevant divisibilities. $\square$

For every unordered pair $p \in \binom{Y}{2}$, define

$$f(s)_p = \frac{a_p \cdot \tilde{s}}{h} \pmod{2}, \quad s \in \Sigma, \quad (13)$$

where $\tilde{s}$ is any integer lift. Let $\Omega = f(\Sigma)$.

Lemma 4.2. *The map in (13) is well-defined and satisfies*

$$f(s+t) = f(s) + f(t), \quad f(cs) = (c \bmod 2)f(s). \quad (14)$$

Consequently Ω is a binary linear subspace.

Proof. The pair constraint is $2a_p \cdot s = 0 \pmod{q}$, so every integer lift has $a_p \cdot \tilde{s}$ divisible by h . For $r = 1$, this is automatic because $h = 1$. Changing the lift by qz changes the quotient by $2a_p \cdot z$, an even integer. The map is therefore well-defined.

Reducing an integer representative of $s + t$ modulo q subtracts qz for some integer vector z ; the same calculation removes this carry after reduction modulo two. Multiplication by a ring scalar behaves identically. Reduction $R \rightarrow \mathbb{F}_2$ is well-defined because q is even, establishing (14). $\square$

4.2 Generating a possibly nonfree kernel

Lemma 4.3. *Given a $b \times m$ matrix B over R , with $b \leq m$, a generating set of $\ker_R B$ with at most m elements can be computed deterministically in polynomial bit complexity in m, r .*

Proof. The case $m = 0$ is immediate. Otherwise, let H be a full-rank row Hermite basis of the integer lattice generated by a lift of B and qI_m . The conditions $Bs = 0 \pmod{q}$ and $HS = 0 \pmod{q}$ are equivalent. Because the row lattice of H contains $q\mathbb{Z}^m$, there is an integer matrix C with $qI_m = CH$. It follows that

$$K = qH^{-1} = C \quad (15)$$

is an integer matrix.

The columns of K , reduced modulo q , generate the kernel. Each column is a solution because $HK = qI_m$. Conversely, for any integer lift of a solution s , write $HS = qz$ with $z \in \mathbb{Z}^m$. Then $s = Kz$, proving completeness. The generators need not be independent over R ; no freeness assumption is made.

Computing H has polynomial bit complexity. Its entries have $O(r+1)$ bits by the bound in Section 3. Exact inversion also has polynomial bit complexity: determinant and cofactor bounds give $O(m(r + \log m))$ bits for the numerators and denominators, and fraction-free elimination computes them in polynomial time. Reduce the resulting integer columns of K modulo q . If B is the output of Algorithm 1, its already computed lattice basis can be reused. $\square$

Theorem 4.4 (Complete image with preimages). *In polynomial bit complexity in n, r , one can compute a basis $\omega_1, \dots, \omega_d$ of Ω , with $d \leq |X|$, and vectors $s_i \in \Sigma$ satisfying $f(s_i) = \omega_i$. Membership in Ω is decidable in polynomial time, with a witnessing multiplicity vector for every member.*

Proof. Apply Lemma 4.3 to obtain generators $g_1, \dots, g_\ell$ of Σ , with $\ell \leq m$. For each g_j and pair p , compute the residue $c_p = a_p \cdot g_j \bmod q$ in $\{0, \dots, q-1\}$. It belongs to $\{0, h\}$ by the pair constraint. Thus $f(g_j)_p = c_p/h$ is an exact integer division; it does not divide by a nonunit in R .

Every $s \in \Sigma$ is an R -linear combination $\sum_j c_j g_j$. By Lemma 4.2,

$$f(s) = \sum_j (c_j \bmod 2) f(g_j). \quad (16)$$

Conversely, every binary combination of the images is the image of the corresponding sum of the generators. Hence these at most m images span all of Ω .

Perform binary Gaussian elimination on them, carrying along their preimages over R . Whenever image rows are added, add the corresponding preimages modulo q . The invariant $f(s_i) = \omega_i$ survives by (14). Remove dependent zero image rows. If $\delta \in \Omega$, solve $\delta = \sum_i \epsilon_i \omega_i$ over $\mathbb{F}_2$ and return $s = \sum_i \epsilon_i s_i \bmod q$.

There are $\binom{|Y|}{2} = O(n^2)$ image coordinates and at most m generators. Image computation uses $O(n^2 m^2)$ elementary arithmetic operations, and elimination with the retained witnesses has polynomial size. Combined with the compression and kernel bounds, this proves polynomial bit complexity. An equation description of Ω may alternatively be obtained from a basis of its binary orthogonal complement. $\square$

Corollary 4.5 (One prescribed-support operation). *A graph H is obtained from G by one valid r -local complementation supported in X if and only if the two graphs agree on pairs meeting X and their edge difference on Y belongs to Ω . The condition is decidable in polynomial bit complexity in n, r , and an accepting instance has constructible binary-encoded multiplicities.*

Proof. Lemma 4.1 identifies all valid multiplicities. For a pair in Y , the toggle predicate is exactly (13). A pair meeting the actual support is unchanged by definition. A pair meeting X outside that support has no common neighbor in X , so its common-neighbor multiplicity is zero and it is unchanged as well. These facts prove necessity and sufficiency. Representatives below q have at most r bits each. $\square$

5 The published graph reduction

This section summarizes the results of Claudet and Perdrix used below. All lemma numbers refer to arXiv:2502.06566v3 [1].

A local set is a nonempty set of the form $D \cup \text{Odd}_G(D)$, where $\text{Odd}_G(D)$ contains the vertices with an odd number of neighbors in D . An MLS cover is a family of inclusion-minimal local sets covering the vertices. The cover assigns each vertex one of the types $X, Y, Z, \perp$, according to the stabilizer operators supported on these sets. We use the polynomial construction in [1], rather than enumerating all minimal local sets.

Standard form has three requirements [1, Definition 3.10]: there are no type- Y vertices; every neighbor v of a type- X vertex u is of type Z and satisfies $u \prec v$ in the fixed vertex order; and every star $\{u\} \cup N(u)$ at a type- X vertex belongs to the chosen MLS cover. In particular, the type- X set is independent. Lemma 3.12 either rejects LU-equivalence soundly or, in time

$O(n^{6.38})$, produces ordinary-LC-equivalent representatives in standard form with respect to a common MLS cover. For LU-equivalent representatives, corresponding vertex types and type- X neighborhoods must agree (Lemma 3.13).

Write X, Z, W for the sets of types $X, Z, \perp$. By Lemma 3.14, surviving standard forms are LC_r -equivalent exactly when one r -local complementation supported in X , together with ordinary complementations at vertices of W , takes one to the other. The two kinds of operations commute because no vertex of W has a neighbor in X . The generalized operation can change only edges within Z .

Lemma 3.15 computes the full space of these edge changes by listing the incidence constraints. Its weights are precisely those in (6). Theorems 3.1 and 4.4 compute the same space in polynomial time. The binary linearity of this space, and its use in the rest of the reduction, are already present in [1].

Given a basis of the toggle space, the construction before Lemma 3.17 deletes X and adds one degree-two auxiliary vertex for each toggled pair in each basis vector. Use the same auxiliary labels in both graphs. Lemma 3.17 reduces the required transformation to ordinary LC on the augmented graphs, subject to two restrictions: no complementation at a vertex of Z , and either all auxiliary vertices belonging to a basis vector occur once or none occur. Lemma 3.18 expresses these restrictions as linear equations on the local-Clifford variables.

The constrained LC algorithm (Proposition 3.5) requires both input graphs to be connected and to have an even-degree vertex. Its cost on graphs of order N , with ℓ additional equations, is $O((N^2 + \ell)N^2)$ binary operations. Section 6 verifies those hypotheses when the toggle space is nonzero, and uses a separate ordinary-LC test when it is zero.

Finally, Corollary 3.29 gives $LU \Rightarrow LC_r$ on graphs with at most $2^{r+3} - 1$ vertices. Every LC_r operation is a local unitary, so this implication gives an equivalence of the two decision problems within that size bound.

6 The full algorithm

6.1 Components and labels

Lemma 6.1. *LU-equivalent labelled graph states have the same partition of their vertex set into connected components. Equivalence then holds if and only if it holds on each corresponding component.*

Proof. A graph state factors across a labelled cut $A : V \setminus A$ if and only if no edge crosses that cut. To see the nontrivial direction, remove the diagonal phases contributed by edges lying entirely within either side of the cut. Up to a common normalization, the bipartite coefficient matrix becomes

$$C_{x,y} = (-1)^{x^T \Gamma_{A, V \setminus A} y}. \quad (17)$$

If a crossing entry of Γ equals one, the rows indexed by the zero vector and the corresponding single-bit vector, and the analogous two columns, give the minor

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (18)$$

It has nonzero determinant. The state therefore has Schmidt rank at least two across the cut. With no crossing edges, the coefficient matrix factors.

Product unitaries preserve factorization across every fixed labelled cut. The finest partition into graph-state product factors is thus the connected-component partition, which must be

the same for both inputs. When partitions coincide, a global product-unitary equality implies equality of the pure-state factors up to component phases, for example by taking reduced states on each component. Conversely, tensoring component transformations supplies a transformation on the whole graph, with their phases absorbed into the global phase. $\square$

Compute these partitions and reject if they differ. One-vertex components are immediate. On a common connected component of order $v \geq 2$, choose

$$r = \max\{1, \lceil \log_2(v+1) \rceil\}. \quad (19)$$

This gives $v \leq 2^{r+3} - 1$ and $r = O(\log v)$, so Corollary 3.29 of [1] makes the LU and LC_r questions identical.

Apply the common-standard-form construction and the type and neighborhood tests from [1, Lemmas 3.12–3.13]. If they succeed, let G_1, G_2 be the standard forms and let X, Z, W be their common type- X , type- Z , and type- $\perp$ sets. The graphs remain connected: any edge removed by ordinary complementation at a vertex is replaced by a two-edge path through that vertex. Compute the image basis from Theorem 4.4 using G_1 and X . Its only possibly nonzero coordinates are pairs in Z , since vertices outside $X \cup Z$ have no neighbor in X . Restricting to these coordinates loses no information.

6.2 The nonzero-image branch

Suppose $d = \dim \Omega > 0$. For each basis vector ω_i and each pair $\{u, w\} \subseteq Z$ on which it is one, add a new vertex p_{uw}^i joined only to u, w , after deleting X . Use the same new labels in both graphs. Let P_i be the set of new vertices for ω_i . Their total number is at most $d \binom{|Z|}{2}$, and

$$N \leq v + d \binom{v}{2} \leq v + |X| \binom{v}{2} = O(v^3). \quad (20)$$

Lemma 6.2 (Applicability of constrained LC). *When $d > 0$, both augmented graphs are connected and have an even-degree vertex.*

Proof. Every basis vector is nonzero, so at least one auxiliary vertex is added; each such vertex has degree two.

For connectivity, fix $x \in X$. The multiplicity vector $h\mathbf{e}_x$ is in Σ , because every weighted incidence row has an even coefficient and hence its scalar product with this vector is divisible by q . Its toggle vector is exactly the clique on $N_{G_1}(x)$. As the basis vectors span Ω , every pair of distinct neighbors of x must occur in at least one basis vector. Some auxiliary vertex therefore supplies a path of length two between each such pair.

Take any two vertices outside X and a simple path between them in G_1 . Independence of X ensures that each passage through an internal vertex $x \in X$ has two neighbors outside X . Replace that passage by the auxiliary path just described. Vertices of X with degree at most one cannot occur internally on the original simple path. Thus the remainder together with the auxiliary vertices is connected. The remainder is nonempty because G_1 is connected, has at least two vertices, and X is independent. All new vertices attach to that remainder. The same argument applies to G_2 , using its connectedness and the identical neighborhoods on X . $\square$

By [1, Lemma 3.17], the required LC_r predicate is now the constrained ordinary-LC predicate on these augmented graphs. In the source's binary local-Clifford coordinates, write the relevant variables as β_u, γ_u to avoid confusion with our incidence matrix B . The equations in [1, Lemma 3.18] are

$$\beta_u = 0 \ (u \in Z), \quad \gamma_p = 0 \ (p \in \bigcup_i P_i), \quad \beta_p = \beta_{p_i} \ (p \in P_i), \quad (21)$$

where p_i is one representative of each nonempty group. There are $O(N + v)$ extra equations. Lemma 6.2 verifies the hypotheses of Proposition 3.5, so the constrained predicate can be decided in polynomial time in the actual augmented size N . Equation (20) transfers that bound to a polynomial in v .

6.3 The zero-image branch

Lemma 6.3. *If $\Omega = \{0\}$, then G_1 and G_2 are LC_r -equivalent if and only if they are ordinarily LC-equivalent.*

Proof. One implication is immediate. For the other, [1, Lemma 3.14] expresses a LC_r equivalence by ordinary complementations on W and one valid generalized operation on X . These operations commute. Since $\Omega = \{0\}$, the latter operation changes no edge of G_1 and can be omitted. Only ordinary complementations remain. $\square$

In this case run the ordinary LC algorithm [3] on the original standard forms G_1, G_2 . Deleting X first could leave disconnected graphs or graphs without an even-degree vertex, outside the constrained routine's hypotheses. Lemma 6.3 supplies a separate exact decision rule, including the case $X = \emptyset$.

6.4 Correctness and bit complexity

Algorithm 2: labelled graph-state LU equivalence

1. Compare labelled connected-component partitions. Reject if they differ.
2. For each common component of order $v \geq 2$, choose r by (19).
3. Use [1, Lemmas 3.12–3.13] to standardize and check types and type- X neighborhoods. Reject on failure.
4. Use Algorithms 1 and Theorem 4.4 to compute the complete toggle basis.
5. If $d = 0$, decide ordinary LC on the original standard forms. If $d > 0$, build the matched gadgets and decide constrained LC using (21).
6. Accept exactly when every component test accepts; singletons always accept.

Proof of Theorem 1.1. Lemma 6.1 justifies the first step and all componentwise decisions. The cited standard-form lemmas make every standardization or type-based rejection sound. On a surviving component, Lemma 4.1 and Theorem 4.4 compute precisely the full toggle space used in [1, Lemma 3.17]. For $d > 0$, [1, Lemmas 3.17–3.18] and Lemma 6.2 make the constrained test equivalent to LC_r equivalence. For $d = 0$, Lemma 6.3 gives that equivalence directly. Corollary 3.29 of [1] and (19) identify it with LU-equivalence. This proves soundness and completeness, and every construction preserves the original labels.

For the complexity bound, the common-cover stage takes $O(v^{6.38})$ time. Compression uses at most $B(r, |V \setminus X|)$ strict enlargements, each represented by a matrix with polynomially many rows and $O(r + 1)$ -bit entries. Its Hermite computations therefore have polynomial bit complexity. Kernel construction, exact image evaluation, and binary elimination have polynomial matrix sizes and bounded residue lengths. Since $r = O(\log v)$ and $2^r \leq 2(v + 1)$, no hidden exponential precision parameter is introduced.

The augmented graphs have $O(v^3)$ vertices. Evaluating the constrained routine's bound on this actual size, rather than on the original graph size, still gives a fixed polynomial. The zero-image branch uses an ordinary polynomial LC test. Summing a fixed polynomial bound

over at most n components remains polynomial in n . We do not optimize the exponent or claim a practical benchmark for the complete procedure. $\square$

7 Exact diagnostics and reproducibility

The accompanying code tests the algebraic constructions and parts of the graph reduction using exact arithmetic. It does not implement the polynomial MLS standardization algorithm, and the finite checks do not prove Theorem 1.1.

7.1 Algebraic checks

The original scripts compare closure with direct subset enumeration on 45 systems, and kernels and images with exhaustive multiplicity enumeration on 36 systems. All passed. However, every original closure example stabilized at its seed lattice. A second script addresses this gap using all Boolean column patterns on $t \in \{3, 4, 5, 6\}$ rows, at six levels. Twelve of its 24 systems require strict enlargement; retaining the seeds alone fails in each of those twelve. The script also checks all 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$, 384 binary incidence systems, and five boundary cases. These records are preserved in the archive.

A fresh implementation uses the dual-lattice formula (15); the earlier implementation uses modular row and column elimination. The new implementation checks every binary 3×3 incidence matrix at levels two and three. It compares the full kernel and image with the literal incidence definition, including the actual support of each multiplicity vector, and verifies every binary combination of the retained image-basis witnesses. This covers 1,024 systems, 294,912 multiplicity vectors, and 2,928 carried witnesses. The dual-lattice method also reproduces every kernel of a 2×2 matrix over $\mathbb{Z}/4\mathbb{Z}$, including 81 kernels of cardinality two or eight, which are nonfree over that ring.

Sixteen additional Boolean-pattern systems test the closure bound at levels up to $r = 256$. All agree with direct subset enumeration, and all attain the bound $B(r, t)$ for their parameters. Two deliberately incorrect variants—omitting triple seeds and omitting the factor two in T_v —give different lattices from the exact construction on the designated control instances.

7.2 Checks of the graph reduction

For connected labelled graphs of orders two through four, we enumerate independent supports and levels one through three. When the toggle space is nonzero, we compare two computations: direct enumeration of the possible toggles followed by graph search over the allowed ordinary complementations, and the auxiliary-graph construction followed by the constrained local-Clifford equations. There are 2,460 comparisons: 1,020 accept and 1,440 reject, with no disagreements. In each case we also enumerate the entire linear solution space of the Clifford equations before checking the determinant conditions. These tests exercise the restricted graph-operation reduction; arbitrary chosen supports need not be type- X sets of a standard form. The 261 zero-image support/level cases are therefore excluded from this comparison, since Lemma 6.3 relies on standard form. A further 1,105 ordinary-LC comparisons agree with unrestricted graph search on small connected graphs satisfying the even-degree hypothesis.

We also use a 27-vertex example to test an operation beyond ordinary LC. Let Z contain six vertices and let X contain one vertex for each four-element or five-element subset of Z , adjacent precisely to that subset. Thus $|X| = 21$. Let G be this bipartite graph, and let H add the clique on Z . Giving every vertex of X multiplicity one yields ten common neighbors for every pair in Z and six for every triple. At level two, all incidence conditions hold and every

pair is toggled. Hence this operation takes G to H .

The stars at vertices of X form a common MLS cover for which both graphs are in standard form. To check this, for each of the 42 stars in the two graphs the code solves for stabilizers supported inside the star and finds only its defining nonidentity generator. The computed toggle space has dimension 15 and contains the clique, with an explicitly recovered multiplicity witness. The constrained gadget test accepts. In contrast, enumerating all 2^8 solutions of the ordinary-LC linear equations finds none satisfying the local determinant conditions, so the ordinary-LC test rejects. This example is a diagnostic control; no novelty is claimed for its construction.

New recheck family	Cases	Outcome
Boolean-pattern systems, $r \leq 256$	16	Exact closure and bound
Dual kernels of 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$	256	All kernels agree
Binary 3×3 incidence systems	1,024	Full kernels and images agree
Carried image-witness combinations	2,928	All valid
Constrained gadget comparisons	2,460	1,020 yes; 1,440 no
Ordinary-LC graph-search comparisons	1,105	379 yes; 726 no
27-vertex higher-level control	1	LU yes; ordinary LC no

Table 1: Exact rechecks in the revised manuscript. Rows describe different objects and should not be added into a single sample count.

All new checks passed in a network-disabled container using Python and SymPy 1.14.0. The archive contains the scripts, full outputs, source locators, evidence map, and unchanged round-21 proof. These are reproducible internal checks, not external mathematical review.

8 Discussion

The improvement comes from the weighted structure of the incidence rows. Closure under coordinatewise multiplication by $2a_v$ replaces subset enumeration, while Hermite reduction keeps at most $|X|$ rows with bounded entries. The subsequent kernel and image calculations preserve every admissible operation, so the rest of the Claudet–Perdrix reduction applies without changing its target space.

The polynomial bound is a worst-case decision result. The $O(n^3)$ auxiliary graph can make a direct implementation expensive, and we have not optimized the resulting exponent. A complete implementation still needs the common-MLS-cover and standardization routines. It would also allow practical comparison with the original subset-based method, which may be faster on small instances. Arbitrary qubit permutations remain outside the labelled decision problem considered here.

Authorship, preparation, and availability

The proof candidate and its exposition were developed with substantial AI assistance under the author’s direction. Two historical internal AI reviews reported no error in the frozen round-21 proof. This revision adds the sharper closure bound (7), the dual-lattice proof of Lemma 4.3, and the rechecks in Section 7. The historical reviews are not reviews of these revisions. Independent expert verification, formal proof-assistant verification, and historical-priority confirmation remain pending.

Scientific Agent Skills [5] guided evidence tracking and scientific writing; it did not verify

the mathematics. The frozen proof and submission are available in [QIQC issue 97](#). The [research page](#) hosts this manuscript and the reproducibility package. The frozen proof was contributed under CC BY 4.0 and the original diagnostic scripts under Apache-2.0; cited third-party works retain their own terms.

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